Cubic Refinements and Euler Characteristics of Outer Space

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Abstract

The stationary law of the local Whitehead chain on cubic multigraphs is proportional to the reciprocal of the full incidence automorphism order. We use cubic refinements to transfer finite sums over all stable rank-r graphs to expectations under this law at $2r-2$ vertices. The refinement multiplicity at a vertex of valence d is $(2d-5)!!$. Applying this identity to the established forest formulas gives distinct observables for the ordinary and virtual Euler characteristics of $\mathrm{Out}(F_r)$. We explain the orientation and stabilizer data required by the forested chain complex, and give exact low-rank checks, variances, and a two-term representative of the first Morita class with an integer dual cocycle certifying its nonvanishing.

1 Introduction

A Whitehead move on a cubic graph collapses an edge and expands the resulting four-valent vertex in a different way. This is also a basic local operation in Outer space, but the finite graph of cubic isomorphism classes records only part of its structure. It forgets markings, edge lengths, forests, and orientation characters. These omissions matter both for homology and for Euler characteristics.

Let $\mathcal{G}_N$ be the connected cubic multigraphs on N vertices, up to isomorphism, with loops and parallel edges allowed. The local chain of [10] chooses an edge and one of its two alternative resolutions, counting moves with multiplicity. A choice of a loop leaves the graph unchanged. Its stationary distribution is

$$\pi_N(G) = \frac{1}{C_N |\operatorname{Aut}(G)|}, \qquad C_N = \sum_{[G]\in\mathcal{G}_N} \frac{1}{|\operatorname{Aut}(G)|}. \tag{1.1}$$

Here and below $\operatorname{Aut}(G)$ acts on vertices and half-edges. Loop flips and permutations of parallel edges belong to this group.

For a fixed rank $r \geq 2$, every connected stable graph is a contraction of a cubic graph on $N = 2r-2$ vertices. Counting those contractions without their multiplicities gives an incorrect sum. Our main identity restores the multiplicities explicitly.

Theorem 1.1 (Refinement transfer)**.** *Let $\mathcal{S}_r$ be the connected rank-r multigraphs with all valences at least three. For $H \in \mathcal{S}_r$ put*

$$R(H) = \prod_{v\in V(H)} (2\deg(v)-5)!!.$$

For every isomorphism-invariant function $u : \mathcal{S}_r \to \mathbb{Q}$,

$$\sum_{[H]\in\mathcal{S}_r} \frac{u(H)}{|\operatorname{Aut}(H)|} = C_N \mathbb{E}_{\pi_N}\left[\sum_{\substack{F\subseteq E(G)\\ F \text{ forest}}} \frac{u(G/F)}{R(G/F)}\right]. \tag{1.2}$$

Each subset of parallel edges is counted separately in the inner sum. Loops cannot belong to a forest.

The proof counts the fibres of the refinement map by orbit–stabilizer [1], using the classical labelled-tree multiplicity [2]. The identity expresses sums over all stable strata on a single fixed cubic state space. We apply it to the ordinary and virtual Euler characteristics and give exact finite checks of the resulting observables.

Culler–Vogtmann introduced Outer space in [8]. Conant–Vogtmann describe the relevant forested graph complexes in [6]. The virtual Euler generating function originates with Smillie–Vogtmann [12]; Borinsky–Vogtmann develop its asymptotics and formal-integral description in [3, 5]. The ordinary Euler characteristic and its generating function are treated separately in [4]. Here we recover these Euler characteristics through cubic refinement transfer. The prism certificate gives a direct verification of the first Morita class in the oriented forest basis of this paper.

2 Cubic chambers and forested cells

2.1 What a move records

A rank-r cubic graph has $2r-2$ vertices and $3r-3$ edges. Positive edge lengths with total length one give a metric simplex of dimension $3r-4$. A marking identifies its fundamental group with F_r. Forgetting both the marking and the lengths gives a state of the finite walk.

The spine of Outer space has dimension $2r-3$. If e is a non-loop edge and a Whitehead move changes G to G', it records a span

$$G \longrightarrow G/e \longleftarrow G'.$$

This span is a path of two edges in the spine. Higher-dimensional cells require several compatible forest collapses. Chamber adjacency alone therefore does not recover the spine's chain complex.

2.2 Orientations and stabilizers

Let $C_k(r)$ be the rational vector space generated by pairs (G, F), where $G \in \mathcal{S}_r$ and F is a forest of k ordered edges. An odd permutation of the forest order changes the sign, and graph isomorphisms identify generators. In particular, a forest whose stabilizer contains an odd permutation represents zero.

For the order $e_0, \dots, e_{k-1}$ define

$$\begin{aligned} \partial_c(G,F) &= \sum_i (-1)^i (G/e_i, F/e_i), \\ \partial_u(G,F) &= \sum_i (-1)^i (G, F\setminus e_i), \qquad \partial = \partial_c - \partial_u. \end{aligned} \tag{2.1}$$

The remaining edge order is inherited. Two successive contractions cancel in opposite orders, as do two unmarkings. A contraction and an unmarking anticommute. Thus

$$\partial_c^2 = \partial_u^2 = \partial_c\partial_u + \partial_u\partial_c = \partial^2 = 0.$$

With finite stabilizers treated in characteristic zero, this is the forested cubical model computing $H_*(\mathrm{Out}(F_r);\mathbb{Q})$ [6, Section 3.2, pp. 1210–1211]; see also [4, Section 2.1]. The presentation by cubic graphs modulo IHX arises from a further filtration; it is not the same generator list.

The full incidence group is essential. If l_v is the number of loops at v and m_{uv} the number of edges between distinct vertices, then

$$|\operatorname{Aut}(G)| = |\operatorname{Aut}_{\mathrm{vertices}}(G)| \prod_v 2^{l_v} l_v! \prod_{u<v} m_{uv}!. \tag{2.2}$$

If a forest uses one edge of a parallel class, the corresponding factor in its stabilizer is divided by m_{uv}, after restricting the vertex group to preserve the marked pairs. A forest uses at most one edge from each parallel class. One may quotient those choices when building a chain basis, but must restore their multiplicities in stabilizer-weighted sums.

At rank two, the rose with two loops has automorphism order eight. The marked theta and dumbbell contract to it with coefficient one in the oriented orbit basis. In the stabilizer pairing $\langle [G,F],[G,F]\rangle = |\operatorname{Aut}(G,F)|$, the adjoint contraction sends the rose to $2\,\theta +$ dumbbell. An ordinary matrix transpose would give coefficients one and one. Move multiplicities and cellular boundary coefficients therefore require different pairings.

3 The refinement identity

The following is the classical count of binary trees with distinct leaf labels. It is also the count underlying phylogenetic tree space [2, Sections 1–2]: root at one of the d labelled leaves to obtain a rooted tree on $d-1$ remaining leaves. We include the short induction to fix the labelling convention.

Proposition 3.1. *A trivalent tree with d distinctly labelled incident leaves has $(2d-5)!!$ isomorphism classes.*

Proof. There is one tree for three leaves. Given a tree with $d-1$ leaves, subdivide any of its $2d-5$ edges and attach the last labelled leaf. Removing that leaf and suppressing the resulting bivalent vertex is the unique inverse operation. Induction gives the formula. □

Proof of Theorem 1.1. Fix a stable graph H. Replace each vertex of valence d by a trivalent tree with its incident half-edges as distinct leaf labels. Proposition 3.1 gives $R(H)$ choices. The inserted tree edges form a forest F in a cubic graph G, and $G/F \cong H$. Conversely every such pair (G,F) determines one of these refinements after choosing an identification of its contraction with H.

The group $\operatorname{Aut}(H)$ acts on the refinement choices, and their orbits are precisely the isomorphism classes of these pairs. The stabilizer of a refinement is $\operatorname{Aut}(G,F)$: an automorphism of a trivalent tree fixing all its incident labels is trivial. Orbit–stabilizer gives

$$\sum_{\substack{[(G,F)]\\ G/F\cong H}} \frac{1}{|\operatorname{Aut}(G,F)|} = \frac{R(H)}{|\operatorname{Aut}(H)|}. \tag{3.1}$$

For each fixed cubic G, summing over its forest orbits with weight $1/|\operatorname{Aut}(G,F)|$ equals summing over all its individual forests with weight $1/|\operatorname{Aut}(G)|$. Multiply (3.1) by $u(H)/R(H)$ and sum over H. Equation (1.1) proves (1.2). All sums are finite. □

The correction can be calculated directly on a cubic graph. A component of F with s vertices contracts to a vertex of valence $3s-2(s-1)=s+2$. Consequently

$$R(G/F)=\prod_{K\in\pi_0(F)}(2|V(K)|-1)!!, \tag{3.2}$$

including isolated vertices as forest components. No graph automorphism calculation is needed for this denominator.

Remark 3.2. The theorem applies to any stable-graph observable, including one which depends on a known automorphism order or orientation character. It does not remove the cost of evaluating that observable. In particular, its use in an expectation does not by itself give a polynomial-time randomized algorithm.

4 The two Euler characteristics

4.1 The virtual invariant

Write

$$f_H(-1)=\sum_{\substack{F\subseteq E(H)\\ F\text{ forest}}}(-1)^{|F|}.$$

The established forest formula is

$$\chi_{\mathrm{virt}}(\mathrm{Out}(F_r))=\sum_{[H]\in\mathcal{S}_r}\frac{f_H(-1)}{|\operatorname{Aut}(H)|}; \tag{4.1}$$

see [5, Section 4]. The sign records cube dimension, while the denominator records its finite stabilizer. For a connected graph the Tutte subset formula gives

$$f_H(-1)=(-1)^{|V(H)|-1}T_H(0,1).$$

If H has a bridge, adding or removing that bridge pairs forests of opposite signs, so $f_H(-1)=0$.

Theorem 1.1 now gives the following complete expectation.

Corollary 4.1. *For $N=2r-2$,*

$$\chi_{\mathrm{virt}}(\mathrm{Out}(F_r))=C_N\mathbb{E}_{\pi_N}X(G),\qquad X(G)=\sum_{F\text{ forest in }G}\frac{f_{G/F}(-1)}{R(G/F)}.$$

By contrast, $-C_N\mathbb{E}_{\pi_N}T_G(0,1)$ gives only the cubic contribution to (4.1). Lower-vertex strata must be restored.

4.2 The ordinary invariant

Let $a_k(H)$ be the number of $\operatorname{Aut}(H)$-orbits of k-edge forests whose stabilizer contains no odd permutation of the forest. Set

$$\epsilon(H)=\sum_k(-1)^k a_k(H). \tag{4.2}$$

The oriented quotient chain complex gives

$$e(\mathrm{Out}(F_r)) = \sum_{[H]\in\mathcal{S}_r} \epsilon(H).$$

This is the alternating sum of rational Betti numbers, called the ordinary or integral Euler characteristic. It is generally different from the virtual Euler characteristic, which is also commonly called the rational Euler characteristic.

For clarity, the character form of Burnside's formula reads

$$\epsilon(H) = \frac{1}{|\operatorname{Aut}(H)|} \sum_{\alpha\in\operatorname{Aut}(H)} \sum_{\substack{F \text{ forest} \\ \alpha F = F}} (-1)^{c(\alpha|_F)}, \tag{4.3}$$

where c counts cycles of the permutation on the forest edges. Indeed the trace of α on the oriented forest basis is $\operatorname{sgn}(\alpha|_F)$. Multiplying by $(-1)^{|F|}$ gives $(-1)^{c(\alpha|_F)}$. Averaging projects to the invariant subspace. This is the formula of [4, Theorem 2.1], grouped here by the underlying graph.

The identity automorphism contribution to the sum over H is (4.1). The other automorphisms supply the difference between the invariants. If parallel choices have first been quotiented out, the identity of the remaining vertex group alone does not recover that contribution.

Apply Theorem 1.1 to $u(H) = |\operatorname{Aut}(H)|\epsilon(H)$.

Corollary 4.2. *For $N = 2r - 2$,*

$$e(\mathrm{Out}(F_r)) = C_N \mathbb{E}_{\pi_N} Y(G), \qquad Y(G) = \sum_{F \text{ forest in } G} \frac{|\operatorname{Aut}(G/F)|\epsilon(G/F)}{R(G/F)}.$$

The factors $|\operatorname{Aut}(G/F)|$ cannot be omitted: the target sum over stable graph types is unweighted.

4.3 Rank two

There are three stable graph types: the rose with two loops, the theta, and the dumbbell. Their automorphism orders are $8, 12, 8$. The respective forest sums are $1, -2, 0$, so

$$\chi_{\mathrm{virt}}(\mathrm{Out}(F_2)) = \frac{1}{8} - \frac{2}{12} = -\frac{1}{24}.$$

The rose has $\epsilon = 1$ and the other two types have $\epsilon = 0$, giving $e(\mathrm{Out}(F_2)) = 1$.

The cubic law has $C_2 = 5/24$ and probabilities $(2/5, 3/5)$ on the theta and dumbbell. The corrected observables are

$$(X(\theta), X(\text{dumbbell})) = (-1, 1/3), \qquad (Y(\theta), Y(\text{dumbbell})) = (8, 8/3).$$

Their expectations give both invariants with the same cubic law. This example detects missing parallel-edge choices, missing loop flips and the confusion between the two Euler characteristics.

5 Exact computations and their scope

5.1 Independent enumeration checks

Every stable graph is reached by repeatedly contracting non-loop edges in a cubic refinement. The programs enumerate those types, deduplicate them, and compute the signed forest sum. For each stable type they separately verify the fibre identity (3.1) using exact rational arithmetic.

An independent check uses the formal Gaussian generating integral of [5]. If Z is a standard real Gaussian, put

$$T(x) = -x - x^2/2 + (1+x)\log(1+x), \qquad \exp \mathcal{Y}(\hbar) = \mathbb{E}\exp\{-\hbar^{-1}T(\sqrt{\hbar}Z)\}.$$

Then $[\hbar^{r-1}]\mathcal{Y} = \chi_{\mathrm{virt}}(\mathrm{Out}(F_r))$. The checker computes the same coefficients from vertex-degree profiles and Gaussian moments, without enumerating graphs or forests.

Rank	Stable types	Cubic types	χ_{virt}	e
2	3	2	$-1/24$	1
3	15	5	$-1/48$	1
4	111	17	$-161/5760$	2
5	1076	71	$-367/5760$	1

These reproduce known values. The virtual fibre check extends through rank six, with 388 cubic classes and 2,746,060 individual forests. The ordinary values are also checked against the alternating dimensions of the oriented chain complex.

5.2 Sampling variance

For independent samples from π_N, the relative variances are:

Rank	$\mathrm{Var}(X)/(\mathbb{E}X)^2$	$\mathrm{Var}(Y)/(\mathbb{E}Y)^2$
2	10.667	0.29630
3	17.052	0.50612
4	20.452	3.96064
5	21.284	74.40849

The observables estimate different targets, so the table does not compare two algorithms for the same invariant. The calculations also omit Markov-chain autocorrelation and the cost of the forest sum. The small-rank values prove no asymptotic variance bound.

5.3 A concrete cycle certificate

The full oriented chain dimensions through rank four are

$$(3,2), \qquad (15,22,9,1), \qquad (111,310,385,312,181,53).$$

Exact rational ranks give $H_0 = \mathbb{Q}$ in ranks two and three and $H_0 = H_4 = \mathbb{Q}$ in rank four, with other groups zero. The separate rank-five certificate also recovers $H_0 = \mathbb{Q}$ and no positive-degree homology. These are established low-rank results.

One compact representative is worth recording. On the triangular prism with triangles 012 and 345 and matching edges $03, 14, 25$, let the ordered forests be

$$A = (01,02,34,35), \qquad B = (01,12,34,35), \qquad z = A - 2B.$$

For a triangle with ordered edges a, b, c, the forest chain $t = a \wedge b - a \wedge c + b \wedge c$ has $\partial_u t = 0$. The product for the two triangles has nine terms. In the oriented prism quotient, three terms give $3A$ and six give $-6B$. The attached certificate verifies

$$\partial_c z = \partial_u z = 0, \qquad \phi\partial_5 = 0, \qquad \phi(z) = 4$$

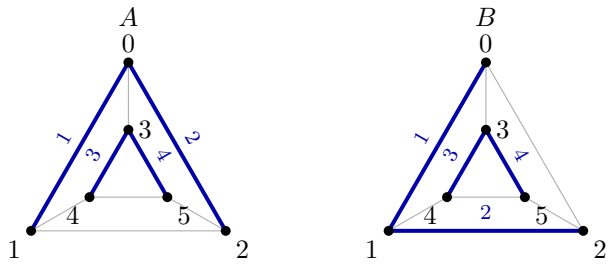


Figure 1: The two ordered forests in the prism representative.

for an integer cochain ϕ with 18 nonzero terms. Thus z is not a boundary. The serialized graphs, edge orders and coefficients allow an independent check without trusting a rank or nullspace search. This represents the first Morita class: the prism is obtained by expanding the vertices of the three-edge theta graph into circles, as in [7, Section 6.1]. The two-term cycle and its 18-term dual cochain form an independently verifiable certificate for this class. The homology dimensions above agree with the known computations summarized in [9]; general nonvanishing of the Morita classes is proved in [11].

6 Reproduction

The accompanying repository contains the source, exact archives and standalone checks in `research/outer-forest-complex`. The commands `make outer-space-paper` and `make outer-space-check` build the manuscript and run the default finite checks. The Python and native graph canonicalizers are compared independently. Rational ranks are computed exactly through rank four. The rank-five archive states separately where modular rank bounds and acyclic quotient arguments certify the rational answer. Agreement modulo two primes alone is not used as a rational-rank certificate.

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